Chelsea Rulangyue Zhao

(647) 646-1366 | chelseazhao9484@gmail.com | Richmond Hill, ON | LinkedIn | GitHub

MEng graduate focused on Data Analytics & Machine Learning, with experience in SQL/Python data preparation, risk classification, model evaluation and Azure-based applications. Former Machine Learning Engineer Intern at CMTE; developed an AI checkout system for Wolseley Express. Available immediately; authorized for full-time Canadian work without current or future sponsorship.

EDUCATION

University of Toronto, M.Eng., Chemical Engineering (Emphasis: Data Analytics & Machine Learning) **Jan 2024 – Oct 2026**

- Grade: A | Selected courses: AI in Finance; Process Data Analytics; Project Management

University of Toronto, B.A.Sc., Chemical Engineering **Sep 2018 – Apr 2023**

- Dean's List Winter 2023 | Minors: Bioengineering; Engineering Business

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | MEng Industry Project **Jan 2026 – Oct 2026**

University of Toronto CMTE | Industry sponsor: Wolseley Express Inc. | Burlington, ON

- Developed a computer vision checkout system for 107 SKUs. Achieved 94.5% complete-checkout accuracy on 2,905 validation images, with all parts and quantities correctly identified without clerk corrections.
- Used SQL to organize, preprocess, analyze and clean incoming data. Built annotation and model-management tools with SAM, DINOv2 and YOLO, including dataset versioning, annotation validation and GPU training/evaluation dashboards.
- Developed object-oriented Python tools with HTTP/REST APIs and Azure Blob/Table Storage; built an optional FastAPI service for Dockerized CPU inference and automated remote GPU training through Linux/SSH/SCP.

Process Technician (Level II) **Apr 2025 – Jan 2026**

Eurofins CDMO Alphora Inc. | Oakville, ON

- Executed 50+ GMP-compliant biopharma operations with audit-ready documentation; collaborated with QA/QC, Engineering and Technology Transfer teams and mentored 7 technicians.

Junior Process Engineer **Jan 2024 – Apr 2025**

Crystal Claire Cosmetics Inc. | Scarborough, ON

- Increased throughput 10% through production scale-up and process validation; reduced a product reject rate from 60% to 5% and trained 20+ technicians.

Process Engineer Co-op **Aug 2021 – Sep 2022**

Atlantic Packaging Products Inc. | Scarborough, ON

- Consolidated production, cost and process data using SQL; prepared and visualized data with Python and maintained DOMO and Excel KPI dashboards. Presented findings to technical and nontechnical stakeholders.

SELECTED PROJECTS

Big Data and AI Case Competition **Nov 2023 – Mar 2024**

University of Toronto, Institute for Management & Innovation (IMI) | Scotiabank | Toronto, ON

- Used SQL to organize, preprocess, analyze and clean all project datasets; built and evaluated XGBoost and LightGBM risk-classification models on 560K financial records, assessing false-negative reduction and imbalanced-class tradeoffs to flag high-risk clients.
- Applied unsupervised anomaly detection, graph-based network analysis, web scraping, and NLP-assisted validation to uncover coordinated entities; Top 5 finalist among 70 teams and featured by U of T IMI.

Teacher-AID | GenAI Genesis Hackathon **Mar 2026**

AI classroom analytics | Two-day hackathon

- Developed a FastAPI backend and retrieval-augmented generation (RAG) with DeepSeek, using student and concept history to generate personalized assessments and teaching recommendations.
- Created synthetic classroom datasets and evaluation dashboards; implemented deployment workflows and tested student- and class-level insights.

ReasonEval | Local LLM Inference and Evaluation **Jan 2026 – Apr 2026**

- Implemented local ReasonEval-7B inference with PyTorch and Hugging Face, including GPU/CPU offloading; evaluated 3,400 solutions from 12 LLMs against human reasoning-error labels across four benchmarks.

TECHNICAL SKILLS

Programming & Data: Python, SQL, pandas, NumPy, Databricks/PySpark, DOMO, Excel; data preparation and visualization
Machine Learning: scikit-learn, XGBoost, LightGBM, PyTorch, YOLO; classification, anomaly detection, model evaluation
Cloud & Deployment: Azure Blob/Table Storage, FastAPI, HTTP/REST APIs, Docker, Git/Azure DevOps, Linux/SSH
Additional Tools: Hugging Face, SAM, DINOv2, RAG, GCP, VS Code